



Data augmentation and hierarchical classification to support the diagnosis of neuropathies based on time series analysis

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ABSTRACT

Diabetic peripheral neuropathy is the most common complication of diabetes, generating physical consequences and limitations in diabetic population. Early diagnosis of this condition based on posture-related tests is still a research issue in the human body movement field. Currently, there is a need for affordable and reliable tools to identify and monitor the progression of this condition. The main goal of this work is to provide a method based on posture feature extraction from the center of pressure time series data, data analysis and machine learning models, in order to find patterns that allow the discrimination among healthy, diabetic and neuropathic people. Time series data from participants are captured using a low cost Wii Balance Board platform, following an experimental protocol. Following a data processing process, that includes reduction of noise, generation of new features, and data augmentation using a method inspired in genetic algorithms to overcome the small number of data. A hierarchical approach is applied in order to deal with class values overlapping. Machine learning algorithms are applied to generate classification models. Results show a range of 89%–97% F1-score, according the case being considered, of the feed-forward neural network models, after the application of the hierarchical approach, being superior to the other classifiers. The high predictive score achieved provides empirical evidence that the proposed approach, based on data augmentation, hierarchical classification and classical machine learning models, can reliably support the detection of diabetic neuropathy condition, also suggesting a simplification of the current procedure.

1. Introduction

Diabetic peripheral neuropathy (DPN) is the most common and costly chronic complication of diabetes. It is a cause of falls and superficial injuries, amputations, disability, and poor quality of life in the diabetic population [1]. DPN is also associated with higher all-cause and cardiovascular mortality in individuals with diabetes [2]. Unfortunately, currently, there is no effective treatment for this condition [3]. Therefore, there is an unmet need for reliable screening and affordable tools to monitor the onset and progression of early neuropathic changes in people with diabetes and facilitate a precise evaluation of future therapies in clinical trials.

In the human body movement field, early diagnosis of DPN based on posture-related measurements is a topic of interest [4–6]. However, previous work has not explored the selection of relevant features to design a model that reliably discriminates the presence of DPN based

on static posture data. In this context, the present work addresses some of the limitations found in the literature, such as the low number of subjects and the absence of weighting multiple features extracted from postural balance to identify the presence of DPN.

In this paper, posture feature extraction from the center of pressure (COP) time series was performed in older persons standing in different sensory conditions. The objective is to find patterns that describe differences in static postural performance between participants with Diabetes, with Diabetes and Neuropathy and Healthy participants. This is carried out through the analysis in the frequency domain, since posturology studies show that the power spectra presents greater differences between healthy subjects and subjects with a health condition, compared to characteristics that are extracted from the time domain.

To obtain COP time series data, the standard approach is to use force platforms. However, a constraint issue in some health services

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maybe its high cost. In this sense, a low-cost alternative is a Nintendo Wii Balance Board (WBB), which has been showed [7,8] that it provides reliable data of body sway during static balance participant trials in the analysis of COP path length. Because of WBB is designed for gaming purposes, it may present overestimation and sample frequency inconsistency that may affect its performance to measure [9,10]. Therefore, the use of the WBB must consider methodological steps in order to minimize these effects. In this regard, an strategy for sample data correction should be performed, for example the SWARII method to apply an interpolation window function that adjusts the values according to the highest concentration by weighting every point with its relevance interval [11].

On the other hand, one of the complexities in these studies is the high cost of participant evaluation, which is reflected in the low number of registered 30 instances. This sample size limitations had been addressed [12,13] and may cause increased risk of Type II errors, less precise estimates, reduced generalization and may cause underfitting in the machine learning algorithms used, mainly when certain classes are overlapped. For overcoming this limitations, data augmentation strategies are proposed and performed to compensate the class imbalance [14,15]. Among this strategies, genetic algorithm strategy, crossover between segments of time series is carried out and, adjusting a series of parameters that assimilate the exchange of genes 35 between chromosomes, new instances are created, inheriting attributes of their respective parents. After verifying the augmented data is consistent with the domain data, the characteristics are extracted and new data will be included in the original dataset to improve the performance of the classification models.

The extracted characteristics allow to discriminate in a greater degree the extreme classes, such as healthy and neuropathic patients. The problem lies in the intermediate class (diabetic), which shares similar characteristics among the previous groups, due to the fact that the pathology gradually worsens. For this reason, the diabetic class tends to overlap with the extreme classes. When entering the dataset with this condition to a classifier, the precision and the recall metrics decline, limiting the accuracy to approximately 70%, which causes a false perception of discrimination between classes.

Another improvement included in this development is the use of an optimizer in the classification scheme: a hierarchical classifier. This function, included in the RapidMiner Studio 9.9 (Technical University of Dortmund, Dortmund, Germany) application, allows the classifier to “follow a path” to classify two classes in each level: healthy with neuropathic and healthy with diabetic subjects. This allows to improve the performance of the classifier in the overlapping class (diabetic), as the classes with the greatest separation are first compared and, at the next level, it is compared with the third overlapped class.

Results obtained show that by including a hierarchical scheme, performance improves. For example, by using two algorithms, a feed-forward network that reaches between 89% and 97% of accuracy and a random forest, that reaches between 71% and 87% of accuracy. This improves the identification of the overlapping class by considering additional validation before labeling new instance. Additionally, data augmentation allows correcting the underfitting problem, by increasing the number of data instances by five times, achieving class balancing. Current dataset is distributed in 20% of healthy, 33% of diabetic and 46% of neuropathic, After data augmentation, new dataset has a proportion of 34% healthy, 36% diabetics and 30% neuropathic of the total data.

Based on the methodology and results of this work, the following contributions are identified:

1. An empirical validation for using a low cost Wii Balance Board (WBB) device to perform an analysis of characteristics extracted from the frequency domain, despite limitations of this type of device.

2. The application of a machine learning approach to the posturology domain, where statistical approaches are frequently used. Several machine learning algorithms are applied, without a previous hypothesis, obtaining several type of classification models, which are different than classical statistical models, where usually a previous hypothesis is required.
3. A data augmentation method for the posturology domain is introduced, based on a genetic algorithm approach, that may be replicated to domains that work with underlying attributes, instead of using raw time series data.
4. A descriptive model designed to improve the classification of the overlapping classes, based on a combination of a hierarchical classification optimizer with a base machine learning algorithm.

The paper is structured as follows: In Section 2, a background is given about the features related to posture and frequency analysis, such as power spectrum and time domain features. In Section 3, a review of the state of the art regarding COP use in related work, the characteristics derived from posture studies, and the use of the functional analysis of time series data is made. In Section 4, data are briefly explored, including the time series, the analysis of the main derived characteristics and their value distributions. In Section 5 materials and methods are explained, including participant's characteristics, the data sensing platform and its meaning, the experimental protocol followed during the tests, data pre-processing, the data augmentation strategy and the machine learning algorithms used for classification. In Section 6, the results obtained from the application of the classification models by classes and by condition of examinations are summarized. Finally, in Section 7, results, limitations and future work issues are identified and discussed.

2. Background

2.1. Study of center of pressure

COP is defined as the application point of all the vertical forces exerted by an individual on a support surface [16]. Sampled values represents the postural sway the body makes to maintain a stable posture. COP enables detecting posture anomalies resulting from diseases or those due to external alterations, such as distractors like dual task or stand on an unstable surface, closing eyes or perform a specific move. To obtain the information, an instrument with sensors that detect the variation of the point where force is exerted is necessary, such as a force platform. However, the cost of these instruments is high and could exceed \$40,000 USD. In [8,17–19], a WBB is used to measure COP in healthy and stroke patients, since the device has a lower cost than laboratory force platforms. Results show that the WBB system can identify differences in balance performance between individuals with stroke and healthy subjects. Therefore, its use is an affordable alternative for static postural behavior analysis and its relationship with diseases.

Through an analysis of the excursion of the COP, in the mediolateral (ML) and anteroposterior (AP) axes, it is feasible to study whether an individual generates greater compensation to maintain a bipedal posture. The displacement in the X coordinate axis corresponds to the ML movement, while the Y coordinate axis corresponds to the AP. A representation of the COP is in Fig. 1, on a human from a front view, in the form of separate time series and its excursion in two dimensions, called stabilogram [20]. This consists of recording postural balance, in an upright position, representing the anteroposterior and medio-lateral swings of the COP. During evaluation of posture in a new patient, several tests are performed mainly in two states: with eyes open (EO) and with eyes closed (EC), since when vision is absent, the brain tries to compensate for the swing, through the proprioceptive system or the somatosensory system [21]. Other distractors like dual tasks and an unstable surfaces are included to increase complexity of sway and

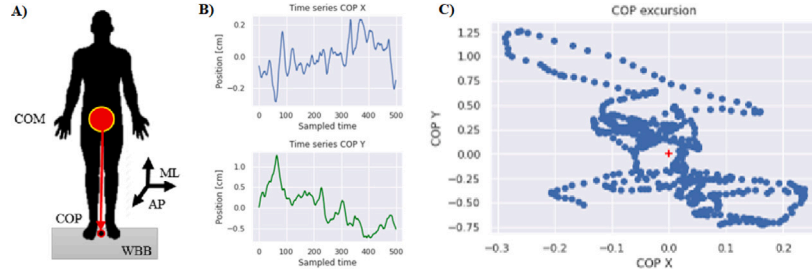


Fig. 1. Representation of a human on the WBB (A), COP in time series format (B) and COP ellipse (C).

reduce the individual's focus of attention. Therefore, unconsciously, the body adapts to this new condition until it reaches a stable position. Similarly, this causes the COP signal to be more natural, since the subject will not be fully aware of how he should position himself in the WBB.

2.2. Frequency analysis features

The work carried out in [22–26] indicates that the greatest information on the health of an individual is found in the Power Spectrum (PS) and attributes derived from posture. PS is used to study the interaction of the three sensory systems involved in postural balance:

1. Vision system, as the primary sensory system used to maintain body balance.
2. Vestibular system and the cerebellum: first is located in the inner ear, which sense the position of the head in relation to the ground, while the second has the function of integrating the sensory information to achieve coordination, precision, and accurate timing.
3. Proprioceptive system: neurons located in muscles, tendons and joints that perceive behaviors in self-movement and body position.

The PS is defined as the Fourier transform of the auto-correlation function [27] of either a continuous or a discrete time series respectively. In the discrete time domain is defined as

$$PS[m] = \sum_{n=1}^N r_{xx}[n] e^{-\frac{j2\pi mn}{N}}; \quad 0 \leq m \leq N, \quad (1)$$

being $j = \sqrt{-1}$ the imaginary unit and $r_{xx}[n]$ its corresponding auto-correlation function. Several transformation methods from the time domain to the frequency domain are applied to get the PS. Since only discrete time series are used in this work, the discrete Fourier transform (DFT)

$$\hat{x}_k = \sum_{n=0}^{N-1} x_n e^{-\frac{j2\pi kn}{N}}; \quad k = 0, \dots, N-1 \quad (2)$$

is applied to calculate the PS of the COP series.

The DFT transforms a sequence of numbers $x_0, x_1, \dots, x_{(N-1)}$, sampled at a frequency k , into a new sequence of complex numbers, say $\hat{x}_0, \hat{x}_1, \dots, \hat{x}_{(N-1)}$. These numbers generate a vector, that complies with the Nyquist Frequency Theorem: the sampled frequency is at least twice the highest frequency contained in the series. Considering that the WBB series are resampled at a rate of 25 Hz (500 samples recorded in a period of 20 s), when applying the DFT, the resulting PS vector is made up of 250 values distributed between the initial frequency 0 and the highest frequency 12. The presence of external random noise is reduced after the preprocessing algorithm, which is properly detailed in Section 5.3 below. This is a relevant point, since the frequency analysis can be sensitive to noise or random fluctuations. Many variants of DFT algorithms are used in the signal processing field. One of them is the Fast Fourier transform (FFT). This is a fast algorithm for

computing DFT using $O(N \log N)$ multiplications and additions [28], allowing to decompose a signal into all of its individual frequency components [29] and its implementation is available in Python libraries such as NumPy [30].

By displaying spectrograms, the distribution of the PS among the 12 frequencies generated by the FFT is analyzed. It allows to identify the interaction of balance systems during the evaluation, mainly, by including disturbances such as the absence of vision, the alteration of attention (through dual tasks) and by making the surface conditions more difficult (instability). These behaviors are mostly perceived in the frequency analysis. Given the limitations of the WBB device, the frequency ranges are limited between 0 Hz and 4 Hz, because the highest concentration of the spectra is found in these ranges. In Fig. 2, PS of body sway COP displacement data set in mediolateral (X) and anteroposterior (Y) axes for eyes open and closed conditions are visualized. Following the line of state-of-the-art works, the frequency ranges are divided into: [0 – 0.5] Hz low-frequency band (LF), associated with the vision system; (0.5 – 2] Hz medium-frequency band (MF), associated with the vestibular system and cerebellum; and (2 – 4] Hz high-frequency band (HF), associated with the proprioceptive system. Visual inspection of Fig. 2 suggests that the healthy (green) and diabetic (blue) groups seem to be separated, only on the X axis, regardless of the visual condition considered. In the case of neuropathic (orange), there is a higher concentration in the low frequencies, in closed eyes conditions. When disturbing vision, the spectra tend to shift to the middle frequencies, being more erratic and greater amplitude than the other classes, indicating greater effort of the vestibular system and the cerebellum to maintain postural balance. At a global level, the trend of the spectra maintains an ordinal behavior, where the amplitude of the PS in diabetic and neuropathic individuals is greater than healthy subjects, as the health of the subject worsens until the development of neuropathy.

Two of the statistical variables most used in these analyzes are mean frequency (MNF) and median frequency. The MNF is the arithmetic mean of the PS, calculated on each band using the FFT. On the other hand, the median is the value in the position that separates the PS in the middle, in their respective bands. These variables seek to generalize the spectrum in a representative value of the subject, which will serve as input for the classifier. Empirically, since the median has a similar distribution to the mean, only the second is used as the attribute. Fig. 3 shows a series of box plots of the MNF between the low, middle and high bands. The separation of the healthy and neuropathic classes is greater in the X axis, in all three bands, while in COP Y only in the MF and HF bands. However, the diabetic class tends to be more complex to differentiate than the neuropathic class. Only on the X axis is there a greater separation of the medians with healthy subjects. But in the case of neuropathic patients, the distribution remains in similar ranges. In the case of the Y axis, neuropathic and diabetic patients differ in that the former present more values that exceed the third quartile, in the MF and HF bands, since the power spectrum tends to be more erratic when distractors are included in the evaluation. This implies that the diabetic class (overlapped) is better separated from the extreme classes if they are compared on their corresponding axes: healthy with diabetics in

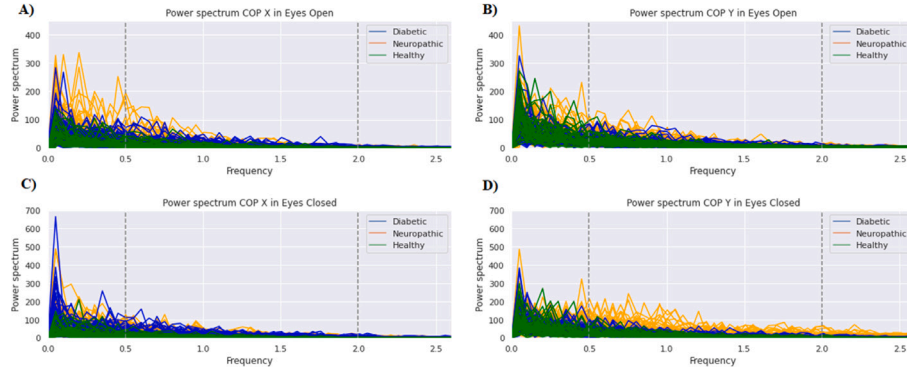


Fig. 2. Power spectrum of COP displacement with eyes closed (C and D) and open (A and B) in mediolateral (COP X) and anteroposterior axes (COP Y).

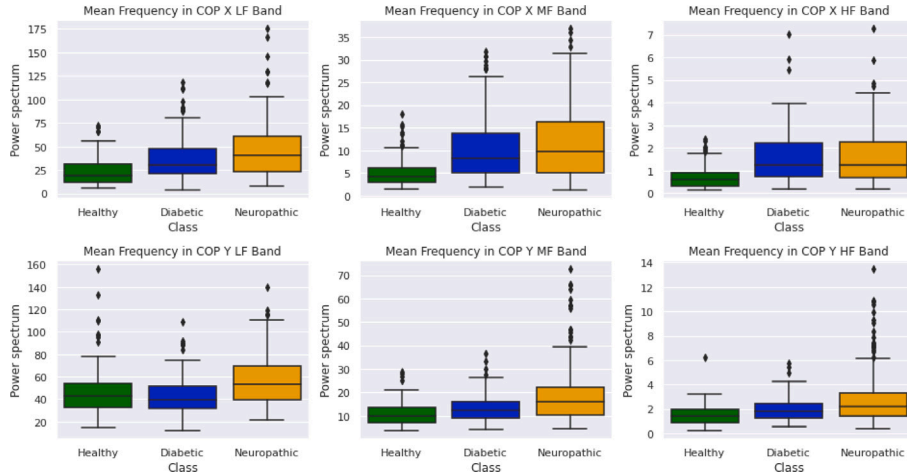


Fig. 3. Box plots of mean PS frequency for COP X and COP Y in all three bands (LF, MF, HF).

COP X and healthy with neuropathic in COP Y but not all three at the same time.

Another attribute used to generalize the spectrum is through the percentage of a part of the total PS. For example, the F50 refers to the median frequency, which divides the PS by 50%. Although the percentage varies depending on the focus of the study, in Vieira's work [18] F80 is used, which refers to 80% of the total power spectrum calculated by FFT. Unlike mean previously calculated in each band, the F80 ranges between the lowest frequency (0 Hz) and the highest frequency (4 Hz), up to 80%, so the summation never exceeds the defined limits. Denoting as N_f the number of different bands (in our particular case $N_f = 4$), the calculation is carried out as shown in Eq. (3):

$$F80 = \sum_{i=1}^{N_f} PS(i) * 0.8. \quad (3)$$

Here PS is a vector that contains the sum of the power spectrum of the N_f frequency bands (with 0 Hz being the lowest and 4 Hz being the highest band).

2.3. Posture features

In posture-based feature extraction related works [8,31,32], time domain attributes are used to improve and find greater differentiation of diabetic subjects. Therefore, three new attributes are considered: Path, Mean Velocity (MeanVel) and Sample Entropy (SampEn). The Path is defined as the absolute length of the movements made by participants in the COP X and Y during the evaluation, and it is

computed as shown in Eq. (4).

$$Path = \sum_{i=0}^{N-1} \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}, \quad (4)$$

where $N+1$ is the number of samples of each time series and $(\Delta x_i, \Delta y_i) = (x_{i+1} - x_i, y_{i+1} - y_i)$ are the increments at the i th step of the COP X and COP Y series respectively, sampled by the WBB.

The mean velocity is defined as the length of the path of COP X and COP Y in a time lapse $T > 0$, and is calculated as shown in Eq. (5).

$$MeanVel = Path/T \quad (5)$$

In our case, all the COP series were truncated in a the time lapse equal to $T = 20$ s.

The Sample Entropy was developed by Richman and Moorman [33] and was designed as an algorithm to evaluate the complexity of a time series. This is also used to evaluate the complexity of the COP displacement. Its interpretation tends to be controversial, as there is still no consensus on whether a value close to 0 is associated with a healthy or pathological condition. In this case, it tends to 1 if the classes correspond to diabetics and neuropathics. Mathematically, it is defined as the conditional probability that two sequences, namely A and B , given a similarity range r , is repeated along the number of data points, N , then

$$SampleEn(m, r, N) = -\ln \frac{A}{B} = -\ln \frac{C^{m+1}(r)}{C^m(r)}, \quad (6)$$

$SampEn(m, r, N)$ is precisely the negative natural logarithm of the conditional probability that two sequences similar form points remain

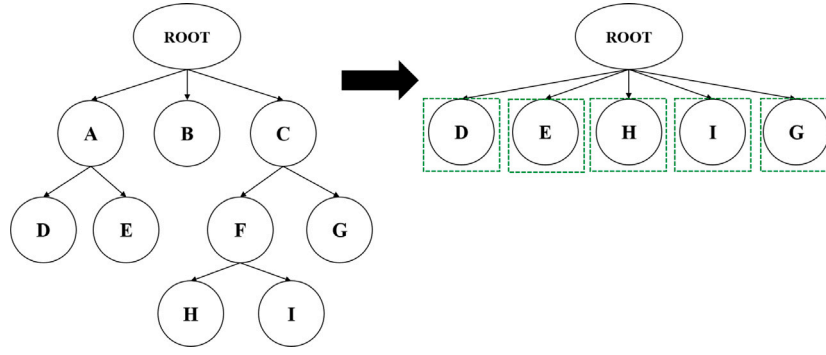


Fig. 4. Flat-hierarchy representation.
Source: Adapted from Silla and Freitas [36].

similar at the next point, where self-matches are not included in calculating the probability. Thus a lower value of *SampEnt* also indicates more self-similarity in the time series.

As shown in Eq. (6), C is the conditional probability that two sequences (C^m the current value and C^{m+1} the next one) of simultaneous points (excluding self-similar patterns), with a window of length m , are similar when tolerance range is less than r . Afterwards, two sequences with $m+1$ elements are taken and evaluated with the m element, and so on, until COP time series of N points is completely considered, being N the number of data points.

2.4. Hierarchical classification

In the study of Silla and Freitas [34], the hierarchical classifier approach is based on the design of a taxonomy, which given a set of classes C , there are subclasses $c_i, c_j, c_k, \dots, c_n$, which can be organized hierarchically, with a node root R , a set of classes that lists the concepts in the application domain, and an “IS-A” relationship, denoted by $<$. Wu et al. [35], describe this type of relationship as asymmetric, anti-reflective and transitive, which meet four conditions:

1. There is a single root R node.
2. $\forall c_i, c_j \in C$, if $c_i < c_j$ then $c_j \not< c_i$.
3. $\forall c_i \in C$, $c_i \not< c_i$.
4. $\forall c_i, c_j, c_k \in C$, if $c_i < c_j$ and $c_j < c_k$ then $c_i < c_k$.

According to [34], the authors describe that hierarchies are categorized depending on the structure of the trees and how classifiers are implemented. The types of hierarchies are:

1. Flat classification approach: the classifier ignores the main hierarchy, focusing only on the attributes of the last leaf of the tree.
2. Local per node: a binary classifier for each node in the tree.
3. Local per parent node: a binary/multiclass classifier for each parent node.
4. Local by level: a binary/multiclass classifier for each level.
5. Global: a single classifier for the entire structure.

A representation of the flat classification approach is found in Fig. 4 and the rest of the approach per node in Fig. 5, where green dotted line represents classification scheme. These taxonomies are independent of the classification algorithm to be used. Also, it will depend on the number of classes the dataset can be grouped into. In Borges's work [37], a competitive neural network is used, where there is an input layer and an output or processing layer, where the taxonomy is found. The model consists of three functions: competition, cooperation, and adaptation. In the competition, Euclidean distance between an instance and a neuron is calculated until the considered winning neuron is found. Then, in the

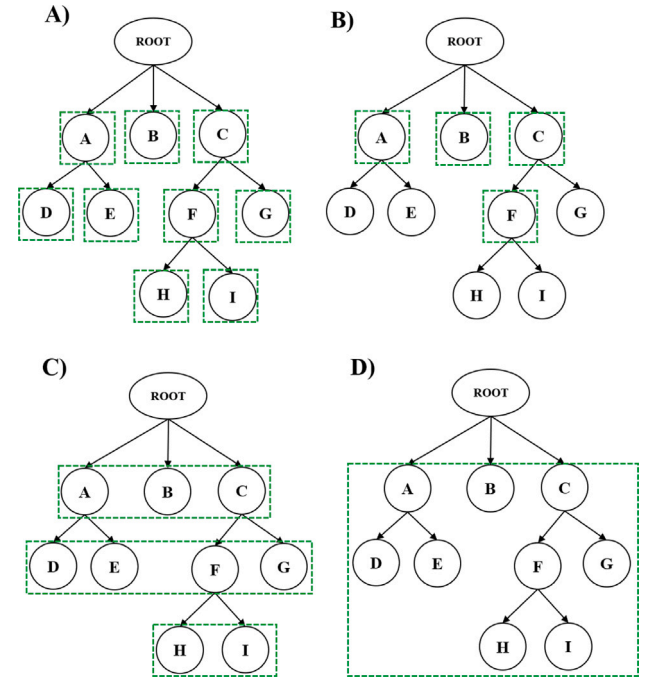


Fig. 5. Local per node (A), local per parent node (B), local by level (C) and global (big-bang) classifier (D) hierarchy.

Source: Adapted from Silla and Freitas [36].

cooperation, the predecessor nodes of the “winner” node are found and finally, in the adaptation, the synaptic weights are updated. This neural network competes with a Naïve Bayes global-model classifier, where a usefulness factor is calculated, so that classes that are deeper in the tree have a higher probability of being classified over those that are more above. These adaptations of classifiers with taxonomies are applied in datasets of protein chains and enzymes, where the number of classes is between 187 and 352.

In the article by Grimaudo [38], the hierarchical classification is used to improve the classification of Internet traffic packets, since one of these can be associated with different protocols and applications: for example, the P2P protocol is subdivided into BitTorrent, BitTorrent MSE/PE, ED2K, etc. Another case is that the HTTP class or protocol is associated with an application, such as Wikipedia, GoogleMaps, or videos such as YouTube, Flash, other, etc. Given the complexity of identifying the traffic packets, it is intended that the flow is classified into few coarse classes. The scheme used is a Local per parent node, with a decision tree algorithm, which is an improvement to the use of a flat classification.

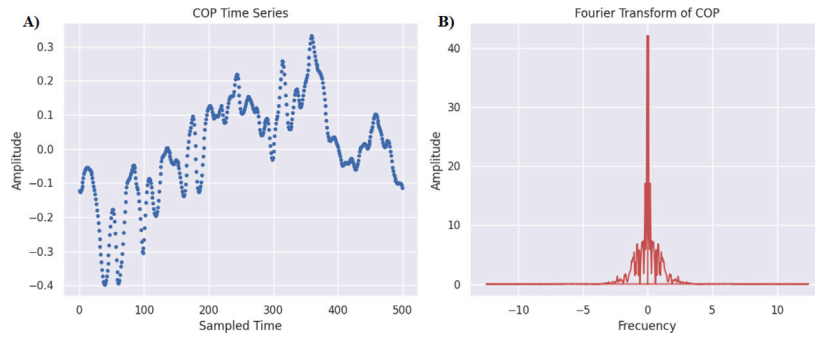


Fig. 6. COP in the time domain (A) and in the frequency domain (B).

In the present case of the posture data, the hierarchy that fits the problem is the Local per parent node, since these structures have a Top-Down processing scheme for each parent node. In the training phase, the hierarchy is processed one level at a time, where there may be one or more classifiers. In the testing phase, each instance is classified in a Top-Down manner, where the new instance is assigned to one or more classes by the first classifier (parent node). Later, at the next level (with another classifier), it assigns one or more “subclasses” of the predicted class (child nodes). The process is repeated until the deepest level of the tree is reached. This function allows to improve the classification process, instructing the algorithm to “follow a path” to identify the instances.

3. State of the art

3.1. Time frequency domain analysis

Frequency is defined as the number of repetitions in which a wave occurs, within a period of time. A signal can be approximated by adding several sine waves, with an appropriate amplitude and phase. In the case of the frequency domain, a complex exponential function is used that allows modeling the original series, generating a visual representation of harmonics, in positive and negative frequencies, according to the function used [39]. Fig. 6 shows the transformation of COP time series obtained from the data set of this research, in the time domain (A) to the frequency domain (B). This represents the power spectrum of the series and its distribution varies depending on the conditions of the evaluated subject.

In the work of Ruth Cabeza-Ruiz [32], balance control is studied in a group of healthy young people with Down syndrome, applying FFT to COP series. Results indicate that, when performing a postural measurement with eyes closed, the power spectrum increases at low frequencies (associated with vision and vestibular systems) in healthy subjects, while it decreases in those affected. On the other hand, the high frequencies (associated with the proprioceptive system or the information that the brain receives from the position and movement of the muscles) are higher in subjects with Down syndrome, in contrast to healthy ones. This highlights the deficiency in the functioning of the balance system, in the absence of vision, by patients with the pathology.

In the work of C. Cruz-Montecinos [22], which compares the balancing strategies between healthy patients and hemophilic arthropathy ones (joint deterioration caused by bleeding) uses frequency analysis, with the aim of identifying which of the systems is affected. The results indicate that, in the ML axis, hemophiliacs present a greater deficiency in the control of sway, in conditions of open and closed eyes: less compensation of the somatosensory system (high frequencies) and greater compensation of balance through the vision (low frequencies). This is reflected in a shift of the power spectrum from right to left: from the mid frequencies to low frequencies.

In the study of Luengas and Toloza [40], PS analysis is performed in a group of subjects who suffered a transtibial amputation, product

of military activities, and who use a prosthesis to replace the affected limb. Unlike previous studies in which FFT is applied in the series, the authors use periograms based on Welch’s method to find the mean and median frequency. Given that amputees present a higher PS, at the three bands, compared to healthy subjects, the authors suggest that the former change their adaptation strategy to balance, exerting greater compensation with the hip and knee muscles of the affected limb.

In [41], a time series analysis of postural tests is performed in order to identifying neuropathies, using statistical techniques and machine learning to classify participants into three categories (healthy, diabetic and neuropathic). This work introduces an underlying probabilistic model, based on the characteristics of the time series, generating average curves for each class. Results of classification models include artificial neural networks, k-nearest neighbors, and naive Bayes machine learning algorithms. In addition, a decision tree model is used to visualize the attribute hierarchy. Besides good maximum accuracy got by machine learning models, it does not include frequency domain features, other posture features derived from the COP, an strategy to increase the number of data that may be generated from observed data, and an approach to deal with the overlapping of the diabetic class with the two other classes, which occurs under several experimental conditions.

Other characteristics derived from the time domain are used together with those from the frequency domain. Both allow us to provide more information about changes in a subject’s posture, as a result of activities or illnesses. In the article by Fraga Vieira [42], attributes such as mean velocity, Root Mean Square (RMS), area of the ellipse, mean frequency, median frequency (F50) and 80% of the frequency (F80) are used to find differences in the posture of hikers due to fatigue. In the work of M. Mancini [39], accelerometers are used in groups of subjects with Parkinson’s disease to obtain the COP. The attributes like RMS, jerk, mean velocity and centroidal frequency are the most reliable measures to differentiate sick subjects from healthy subjects.

Although previous works list a series of extracted features that facilitate the identification of groups of subjects, none of the previous works use machine learning algorithms, due to the high cost of evaluating subjects. Approximately the subjects do not exceed the number of 20 participants. For this reason, these studies focus mainly on statistical analyses. The advantage of this research is to use machine learning classification algorithms on those attributes that are most repeated.

3.2. Data augmentation in time series

Data augmentation corresponds to all techniques applied to create new synthetic or artificial data, from an original dataset, through the use of machine learning algorithms, statistical models or manual transformations [43]. The low number of instances in a data set can lead to biases in the algorithm used. For example, if a dataset contains 70% or more tuples of a class A and the remaining 30% or less of a class B, then classification algorithm tends to overfit, so that most new instances could be classified as A.

The author Forestier [44] proposes to use the Dynamic Time Warping algorithm to generate an average curve of the original series. Then, from this, generate replicas of series, assigning weights to each value, using an Expectation-Maximization strategy: first, multiple series are aligned around the original average curve and, second, the average curve is updated based on the new weights. This generates new synthetic series with a similar structure, unlike the edges, which cause a displacement, depending on the tolerance range.

In the work of C. Oh [45], the interpolation method is used, in which unknown values are estimated from the original data. This algorithm uses cubic spline interpolation, with a third degree polynomial $P(x) = ax^3 + bx^2 + cx + d$. Additionally, a “data collection” factor indicates what proportion of the original series will be resampled, with the same frequency as the original. This allows to increase the number of observations in the series (for example, resample from 500 to 5000 samples), without losing the original structure.

Another strategy used are based on genetic algorithms. These are based on a “population” of data evolving, simulating evolution by natural selection and using the principles of variation, selection and inheritance [46]. In the work of Nethravathi [47], which data augmentation is carried out by applying the conventional processes of genetics: selection, crossover, mutation and validation. The augmentation is made on a dataset that collects characteristics of customers, based on their lifestyle, environment, activities, hobby, etc. In this way, for each tuple the columns are assimilated as a chromosome, where with a random function, they are selected and the crossover and mutation are carried out, with other tuples. Subsequently, the validation process makes it possible to discard inconsistent data, generated by the crossover effect. An adaptation of this model is applied in COP time series. All time series must be divided into a number of segments, adjusted empirically until to find the optimal combination that generates the least inconsistent or duplicated series.

3.3. Time series prediction based on deep learning

Recently, Deep learning models are being used in several industrial and academic research domains, based on notable results in image classification, natural language processing and reinforcement learning applications [48–50]. In relation to time series analysis, in [51] a survey is performed on time-series forecasting based on deep learning models, in particular describing how temporal information is used to make one-step-ahead and multi-horizon time series predictions, highlighting recent advances in hybrid deep learning models which combine statistical and neural network models to improve predictive results.

In [52], an adaptive fused spatial-temporal graph convolutional network is introduced to deal with the unknown spatial-temporal structure problem for multivariate time series (MTS) prediction. This proposed network structure extracts hierarchical feature information to enhance the predicted accuracy in multivariate time series. Experiments over several datasets show that generated models get state-of-the-art performance when compared with other deep learning models in the tasks of single-step and multi-step prediction over multivariate time series. In the same direction, in [53], based on convolution neural network (CNN) and attention mechanism [54], a convolution LSTM network model with two-stage attention for multivariate time series prediction is proposed. In one hand, it allows the extraction of spatial correlation of MTS and, on the other hand, the LSTM model extracts temporal correlation. The application of these network models on three different domains achieve state-of-the-art performance when compared to other four deep learning models, being suitable for single-step and multi-step prediction of MTS.

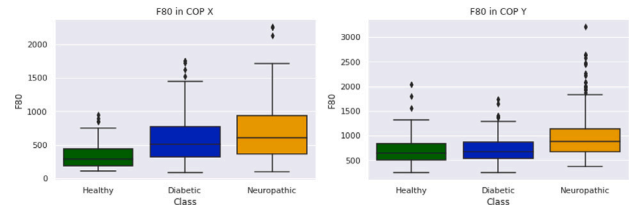


Fig. 7. Box plot of F80 in COP X and COP Y power spectrum.

4. Data exploration

Fig. 7 shows the calculation of the F80 on the X and Y axes. Following the evidence found in the calculation of the MNF, healthy group has a lower spectrum than the diabetic and neuropathic subjects, which means better posture control and therefore a less erratic PSD. In the case of COP Y, neuropathic patients have a greater power spectrum than the rest, which considers F80 as a candidate attribute to improve the classification model. However, the complexity is maintained to differentiate the overlapping class (diabetics). This does it necessary to include other attributes derived from posture.

Fig. 8 shows the box plot of each posture attributes, downwards, in both axes (A) and separated in COP X (B) and COP Y (C). Empirically, by incorporating the age of the subject and performing a filter between ranges, a greater separation of the three classes of subjects is obtained, mainly on the Y axis, in the range less than or equal to 60 years old. However, the diabetic class tends to overlap between healthy and neuropathic patients, similar to what was reviewed in the frequency domain. This occurs most notably in MeanVel and SampEn (rows 1 and 2, respectively). In the case of the X axis, it is not feasible to separate them from the neuropathic ones, but only with healthy subjects.

In the case of Fig. 9, box plot corresponds to subjects who are older than 60 years old. MeanVel, SampEn and Path, on the X axis, have a growing trend, increasing from healthy to neuropathic, as the disease worsens. However, on the Y axis, the behavior of healthy people tends to resemble to diabetics. Due to these changes, that is, including the axis as an additional attribute and performing separation between age ranges, the classifier performance improves.

5. Materials and methods

5.1. Participants

Data were obtained from the Department of Kinesiology at Universidad de Antofagasta. The evaluated participants are thirty-two seniors, 17 men and 15 women, with a mean age of 64 ± 9 years old. Participants are classified into three groups: healthy group (HG; $n = 11$), diabetic group (DG; $n = 9$) and diabetic with diagnosed diabetic neuropathy group (DNG; $n = 12$). Inclusion criteria for DG were: Diabetes diagnosed over two years ago and independence in carrying out life activities according to Barthel Index Score. Participants were excluded if met any of the following exclusion criteria: Neurological conditions other than DN, vestibular disorders, uncontrolled hypertension, cognitive impairment according to the Minimental test, vision impairment, hypoglycemia, amputation, or any traumatic lower limb injuries in the past six months. Class labeling was performed by an evaluator using four standardized tests: the Barthel index [35], the Neurological Symptom Score (NSS) [37], the Diabetic Neuropathy Symptom (DNS) [38] and the tuning fork vibratory sensitivity test in 128 Hz and 256 Hz (in lower extremities) [55]. Participants demographics are shown in Table 1. All subjects gave informed consent to participate in the study. The experimental procedures were approved by the University of Antofagasta Ethics Committee, (CEIC-UA, Number 155/2018) and all requirements set out in the Declaration of Helsinki (1975, subsequently revised in 2008) were also met.

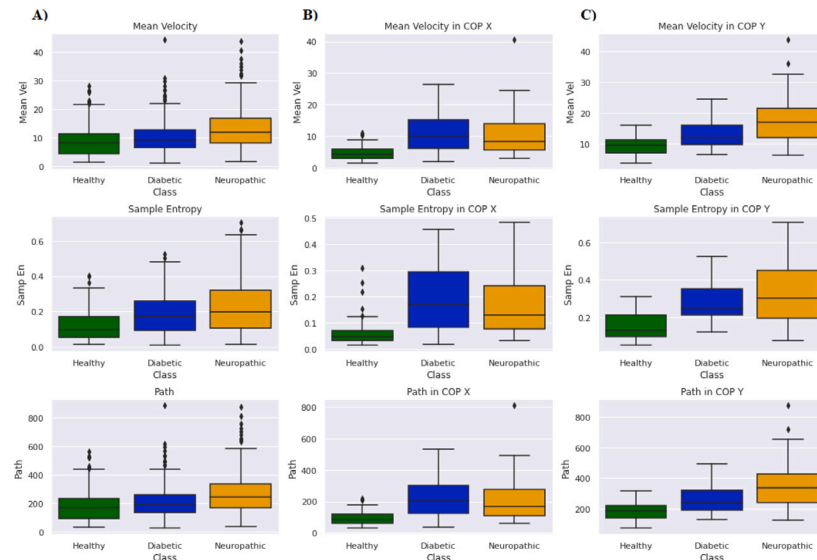


Fig. 8. Box plot of MeanVel, SampEn and Path distribution in both axis (A), in COP X (B) and COP Y (C), on subjects with less than or equal to 60 years old.

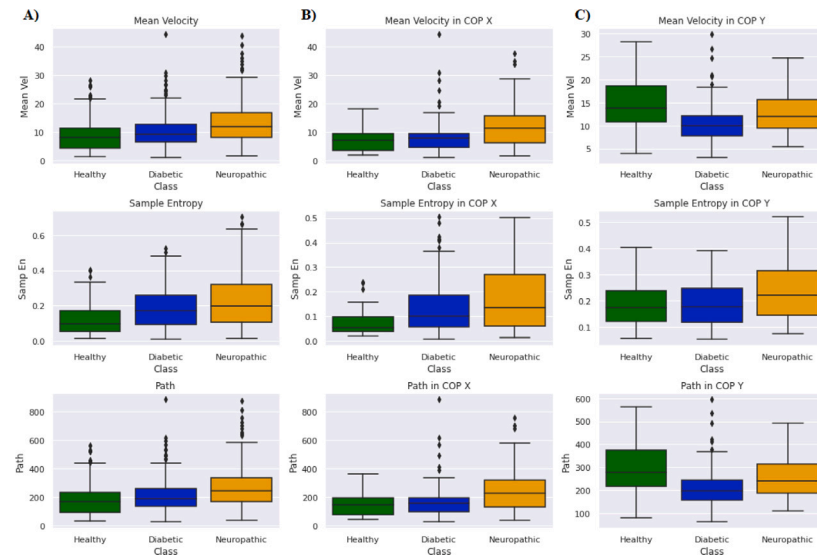


Fig. 9. Box plot of SampEn, MeanVel and Path distribution in both axis (A), in COP X (B) and COP Y (C), on subjects older than 60 years old.

Table 1
Demographic participant data.

Variable	Healthy (n = 11)	Diabetes (n = 9)	Neuropathy (n = 12)
Age, years	60.27 ± 8.60	65.67 ± 11.10	66.13 ± 6.79
Weight, Kg	73.77 ± 10.60	79.61 ± 7.84	68.44 ± 18.57
Height, m	1.62 ± 0.07	1.62 ± 0.08	1.61 ± 0.09
BMI, Kg/m ²	28.03 ± 3.88	30.32 ± 2.59	26.24 ± 6.31
Diabetes diagnosis, years	—	21.89 ± 11.06	18.38 ± 13.94
Glucose, mg/dl	99.09 ± 13.44	149.78 ± 55.33	207.50 ± 82.01

5.2. Experimental protocol

An instrumented version of the modified Clinical Test of Sensory Interaction for Balance was applied [56]. An unstable surface and dual task were used to test postural control in four increasingly challenged balance conditions. The first consisted of a low density foam surface that was positioned on the WBB. The second one consisted in a backward count from a random number indicated by an evaluator, in order to alter the participant's focus in the test. Subjects were positioned on the WBB in an upright and still posture, with their arms relaxed and

quiet, while the physical therapist performed the test, in an estimated period of 30 s. In total, eight different test conditions were performed by participants:

- Eyes open on stable surface.
- Eyes closed on stable surface.
- Eyes open on unstable surface.
- Eyes closed on unstable surface.
- Eyes open on stable surface with dual task.
- Eyes closed on stable surface with dual task.

- Eyes open on unstable surface with dual task.
- Eyes closed on unstable surface with dual task.

To increase the amount of data, the evaluator repeated the entire procedure up to three times, depending on the conditions of each patient.

5.3. Data preprocessing

WBB contains four sensors in each corner: upper left (UL), upper right (UR), lower left (LL), and lower right (LR), which are used for calculating COP. Assuming that the separation distance between the sensors from left to right x is 433 mm and that of the upper and lower sensors y is 238 mm [57], then, COP calculation at X and Y axis respectively is computed as follows

$$COP_x = \frac{x}{2} \frac{(UR + LR) - (UL + LL)}{(UR + LR + UL + LL)} \quad (7)$$

$$COP_y = \frac{y}{2} \frac{(UL + UR) - (LL + LR)}{(UR + LR + UL + LL)} \quad (8)$$

When a subject is positioned on WBB, this starts to measure his COP excursion on both axes. The unit data is expressed in kilograms. So, the sum of the four sensors determines the subject's total weight. During evaluation, data are transmitted to a lab computer, through BrainBloX software (University of Colorado, Boulder, Colorado, United States), which communicates both devices via Bluetooth. Finally, data are stored in the lab computer. The patient's class is not defined by the device, but rather by the clinical evaluation of the physical therapist expert, so that data can be preprocessed, labeling the time series with a class, according to the physiotherapist's evaluations.

WBB mean frequency is 63 Hz; however, it shows differences, according to three regimes: fast-paced (> 100 Hz), moderate (≈ 54 Hz), and slow (≈ 35 Hz). Due to this, the method known as Sliding Window Average with Relevance Interval Interpolation (SWARII) is used to correct the problems of inconsistent data sampling [55], by interpolating data adjusting the values according to their highest concentration. More precisely

$$X_{SWARII}(s) = \frac{1}{\Delta} \sum_{t \in T_s(\Delta)} x_t I_{T_s(\Delta)}(t), \quad (9)$$

where I is the Relevance Interval (RI), s is the current instant of time and $T_s(\Delta)$ is a time window of length $\Delta > 0$ centered in s . Then each observation x_t of the signal is fitted with the value of I . The value of RI will depend on the separation of the current value and the rest. Average windowing reduces signal noise. The main idea of the algorithm is that if there are outliers, the average window controls the amount of smoothing of the estimation, which reduces the loss of information in the resampled data. The SWARII algorithm is applied to all the time series previously obtained from the 30-second experimental protocol. Next, all the signals are resampled to 25 Hz. After this, all time series are truncated in 1.5 s at the start, then 20 s (500 samples) are selected and the remaining samples are removed. Series normalization includes subtracting the mean value of position to center the signal on the X and Y axes.

5.4. Dataset design

The design of the data set for the present study was carried out in several stages:

- Data collection: initially, COP time series in the X and Y axes of the participants were obtained, from the experimental protocol and using the Brainblox software. These time series were stored in CSV format for further processing and analysis. In total, 433 files were collected (COP X and COP Y included in one file), distributed in 87 for HG, 145 for DG and 201 for DNG.

- Data preprocessing: the time series obtained were subjected to the SWARII algorithm, designed specifically for time series from the WBB. This algorithm aims to remove inherent noise and resample the time series, thus ensuring data quality for subsequent analysis.
- Feature extraction: after preprocessing, features were extracted from the time series: Path, MeanVel, SampEn, mean PS in LF, MF and HF bands and F80. This extraction was carried out on each series individually. In addition, three additional attributes were incorporated into the data set: the axis to which the COP corresponds (X or Y), the sex and age of the subject.
- Class labeling: the physical therapist is the expert who carries out the labeling of the subject during the experimental protocol. Therefore, each instance of the extracted feature set was labeled with the class to which COP X and Y belonged. These classes can be HG, DG, or DNG, depending on the clinical and diagnostic characteristics of the individual.
- Analysis and validation: the primary method used to ensure data quality and consistency was data visualization using boxplots. These plots provide a clear graphical representation of the distribution of the features extracted, allowing a visual evaluation of possible outliers, dispersion and central tendencies. The boxplots revealed a characteristic distribution across the quartiles of the extracted features, where the lowest values were typically associated with HG, the overlapping mean values corresponded to DG, and the highest values to DNG.

The results obtained suggest that the data collected is adequate, providing a data set of 866 instances, which will be used for data augmentation.

5.5. Data augmentation

To compensate the problem of data imbalance and improve the classifier performance, a data augmentation raised by Nethravathi [47] is used. A genetic algorithm method is designed to perform random crossing between segments of time series of the same class. To implement this algorithm, Python 3 (Python Software Foundation) from the Google Colab tool (Google LLC, Mountain View, California, United States) is used, an online platform for running Jupyter notebooks. The functions used to generate new series are:

1. Selection: two groups of four candidates each, without repeating, are selected randomly, then only one of each group is selected. Each selected individual, say I_1 and I_2 , carry out its COP X and COP Y information, among other information. The selection is always made between time series of the same class. That is, a series from a healthy subject is never selected with that of a neuropathic subject. Same with axis. Fig. 10 represents the result of the selection process of two COP time series (COP 1 and COP 2).
2. Crossover: selected time series are divided into 10 segments and I_1 is crossed with the k segments of I_2 , randomly. Empirically, it is established that: $k_{healthy} = 5$, $k_{diabetic} = 5$, $k_{neuropathic} = 4$, being the candidate I_1 the one that inherits the clinical attributes of the patient (age, sex and test). Fig. 11 represents the result of the crossover process. The new time series ($R1$) now has the ten segments merged from I_1 and I_2 .
3. Mutation: similar to crossover, series are divided into 10 segments, but four of them are randomly selected as candidates to mutate, calculating the mean between the crossed COP X R_1 and I_2 , in segment k . Fig. 12 represents the result of the mutation process. The last COP time series generated R_2 will be a candidate to be a new augmented data.
4. Validation: activities of elimination of duplicates, comparison of the PSD and the attributes derived from the position are analyzed, between the artificial and real data, in order to remove inconsistencies. Fig. 13 shows the final COP series generated.

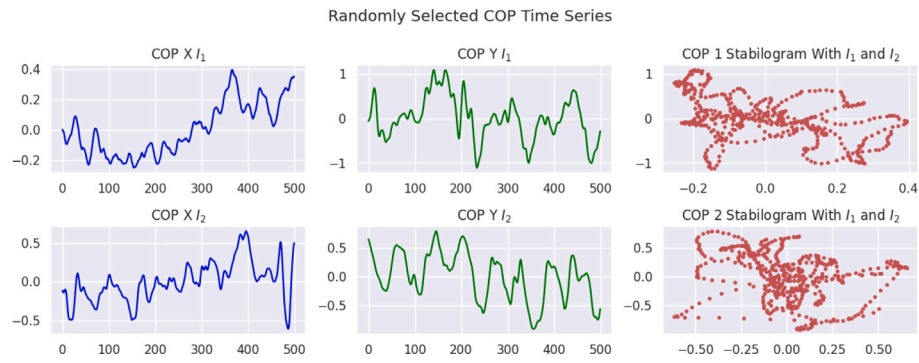


Fig. 10. Example of the selection of two COP X and COP Y with their respective stabilograms.

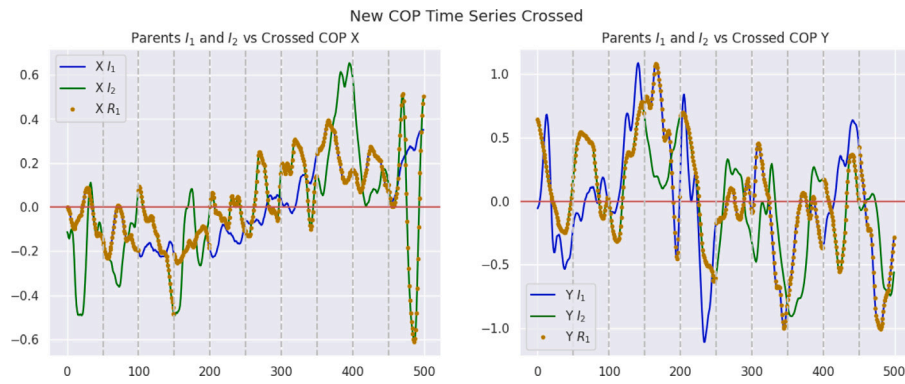


Fig. 11. Crossover process between I_1 and I_2 with the new COP R_1 generated.

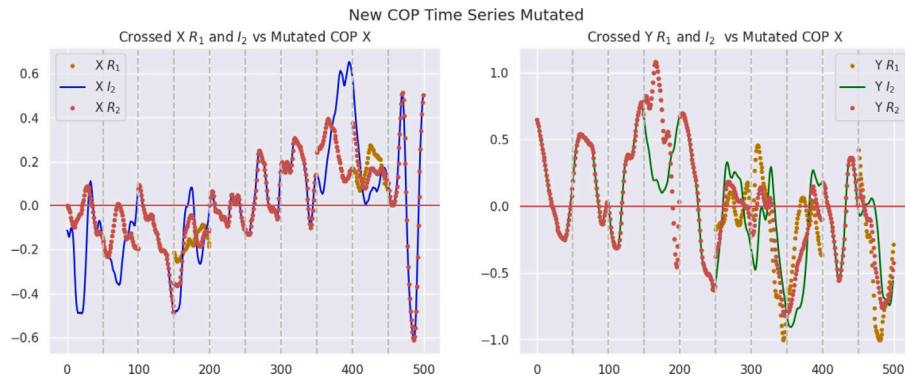


Fig. 12. Mutation process between R_1 and I_2 with the final COP R_2 generated.

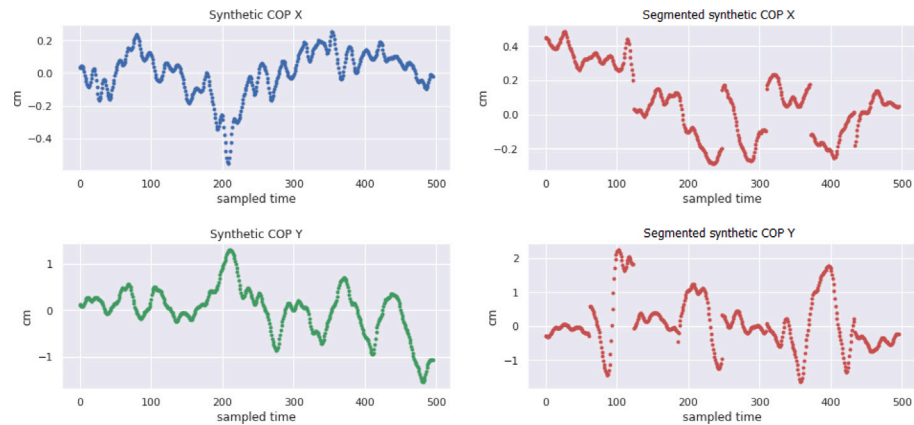


Fig. 13. The final augmented time series using the genetic algorithm.

Table 2

Summary of genetic algorithm criteria for data augmentation.

Class	Crossover	Mutation	Median path	Median F80	Selected instances
Healthy	10/5	10/4	95.93	633.48	1,614
Diabetic	10/5	10/4	164.05	670.68	1,592
Neuropathic	10/4	10/4	209.45	887.44	1,192

This process is executed with 5000 iterations for each axis and its respective class. After removing duplicates, it is considered to generate the same number of instances for each class and axis, with a lower bound of 4950 series per axis, which gives a total of 9900 per class, that is, 29,700 artificial time series. A final selection criterion is performed to select the data, so that 4398 artificial series are chosen, which is six times the size of the original 866 series. The final selection criterion is performed by analyzing the medians of the two attributes in the original dataset: Path and F80. The first allows to separate the three classes, on the X axis, while the second performs it on the Y axis. Later, instances closest to the median are randomly selected, in order to avoid increasing of class overlap. However, this selection does not prevent certain outliers from being included in the final dataset. A summary of this is shown in Table 2. Finally, the consolidated dataset has 5264 instances, 16% corresponding to real data and the remaining 84% to artificial data, distributed in: 34% for healthy, 36% for diabetics and 30% for neuropathic subjects. In each model, the dataset is separated into 70% for training-validation and the remaining 30% for testing. Empirically, Cross-Validation function obtains the highest performance if the dataset is divided into 20 folds, leaving the last one as the validation set.

Algorithm 1 Genetic Algorithm for Data Augmentation COP Time Series

```

1: function GENETIC_ALGORITHM(dataset, Axis, Class)
2:   dataset ← Filter dataset by Class and by Axis
3:   medianF80 ← Get_Median_Value(dataset)
4:   medianPath ← Get_Median_Value(dataset)
5:   new_dataset ← ∅ # New dataset for augmented data
6:   for iterations = 1 to 5000 do
7:     if Axis == x then
8:       I1, I2 ← selection(dataset, medianPath)
9:     else
10:      I1, I2 ← selection(dataset, medianF80)
11:    end if
12:    R1 ← Crossover(I1, I2)
13:    R2 ← Mutation(R1, I2)
14:    R2 ← Validation(R2)
15:    if R2 <> null then
16:      new_dataset ← new_dataset ∪ R2
17:    end if
18:  end for
19:  new_dataset ← Remove duplicate series
20:  if Class == HG then
21:    Select 1614 series randomly
22:  else
23:    if Class == DG then
24:      Select 1592 series randomly
25:    else
26:      if Class == DNG then
27:        Select 1192 series randomly
28:      end if
29:    end if
30:  end if
31:  return new_dataset
32: end function

```

5.6. Machine learning algorithms

The Rapidminer Studio 9.9 software is used to design the algorithms. This software allows for the rapid design of machine learning

Algorithm 2 Functions of Genetic Algorithm

```

1: function GET_MEDIAN_VALUE(population)
2:   Get median value from attribute
3:   return median
4: end function
5: function SELECTION(population, median_value)
6:   I1 ← Select candidates randomly from each group closest to median_value
7:   I2 ← Select candidates randomly from each group closest to median_value
8:   return I1, I2
9: end function
10: function CROSSOVER(I1, I2)
11:   segments ← 10
12:   k ← 4
13:   for j = 1 to segments do
14:     R1 ← Cross I1 and I2 randomly without repetition
15:   end for
16:   return R1
17: end function
18: function MUTATION(R1, I2)
19:   segments ← 10
20:   if Class == Neuropathic then
21:     k ← 4
22:   else
23:     k ← 5
24:   end if
25:   for j = 1 to segments do
26:     R2 ← Mutate R1 and I2 without repetition
27:   end for
28:   return R2
29: end function
30: function VALIDATION(R2)
31:   Evaluate R2 using PSD, Path and F80 with population
32:   if R2 is outlier then
33:     R2 ← null
34:   end if
35:   return R2
36: end function

```

models, easy configuration of hyperparameters and agile testing with the generated data sets. In addition, Python was used to display graphics and support the design of the datasets. This software is installed in an Asus GL553VE computer, with a 2.8 GHz Intel Core i7-7700HQ Quad-Core processor, NVIDIA GeForce GTX 1050Ti (4 GB GDDR5) video card, 16 GB of DDR4 RAM and a 256 GB SSD hard drive, with operating system Windows 10 Home (64-Bit). Four models were selected for classification: Naive Bayes (NB), Feed-Forward Neural Networks (FNN), K-Nearest neighbor (K-NN) and Random Forest (RF). The NB classifier is based on Bayes' theorem, in which, given a set of training tuples labeled with their respective class, the conditional probabilities are calculated and the class with the highest value is selected. Each tuple consists of a set of attributes $X = (x_1, x_2, x_3, \dots, x_n)$. If each of these belongs to one of the three classes of patients, then the classifier predicts the group it belongs to, evaluating the maximum a posteriori hypothesis given the X vector. The model considers that there is a conditional independence among attributes, which reduces the computational cost when processing a large dataset.

The FNN model consists of a set of interconnected nodes separated into an input layer, hidden layers, and an output layer. The attributes are separately entered in parallel, and as they go through them, the values are fitted by adding synaptic weights. These are adjusted iteratively until the least mean square error is found. The updating of weights is carried out by backpropagation, by adding the error obtained from the input unit and the respective weights, multiplied by a learning rate, which goes from 0 to 1. The function that allows improving the search for the minimum error is known as gradient descent. Over

Table 3

Summary of parameters configured in the FFN network.
Source: Extracted from Rapidminer.

Parameter	Value
Activation function	ReLU
Hidden layers	2 of 50 neurons each
Epochs	30
Adaptive learning rate	True
Optimizer	SGD
Regularization	L1
Loss function	Cross-Entropy

Table 4

Summary of parameters configured in the RF algorithm.
Source: Extracted from Rapidminer.

Parameter	Value
Trees	10
Criterion	Gain information
Maximum depth level	6
Apply pruning	True

time, improvements to neural networks have been incorporated, such as optimization functions to reduce error. One of these is Stochastic Gradient Descent (SGD), in which training instances are selected based on batch size, rather than using the full dataset. In outputs, a Rectified Linear Unit (ReLU) activation function is used to adjust non-linear regressions to achieve the activation or inhibition of subsequent neurons. When the problem is a multiclass classification, the output function used is a Softmax. This is a generalization of the sigmoid function and differs in that the output vector corresponds to the probabilities that an instance belongs to a class. The sum of these is equal to one, being the corresponding class the highest value. A summary of parameters is shown in Table 3.

The K-NN model is used mainly in the area of pattern recognition and is based on similar learning tuples. The model allows searching for the pattern space of the k unknown values that are closest to the training values. The degree of “closeness” between two tuples is based on the Euclidean distance between them. When a new value is evaluated, the majority class of the closest k -neighbors is assigned.

Decision trees are graph architectures, which the information gain between attributes is obtained. This model shows the hierarchy of attributes that best classifies the examples. First, the entropy is calculated by the probabilities of each one of its values in the dataset. Then, for each value of the attribute, the probability is calculated with each value of the class attribute. The information gain of an attribute is the difference between the entropy of a variable and the sum of the entropy of the rest and its class. Another advantage of this model is that it allows “pruning” the branches, when a tree extends over several levels. This supports the expert to understand which attributes are the most relevant and which ones do not improve accuracy. A method that improves the traditional decision tree is the Random Forest (RF), which considers an ensemble of tree classifiers (hence its name). This scheme corresponds to trees that are generated using a random selection of attributes, to perform the division through entropy. Depending on the algorithm configuration, the number and distribution of nodes for each returned tree varies. Therefore, it is not considered as a model to define a single hierarchy, but as a way to force the algorithm to improve the division of attributes. A summary of parameters is shown in Table 4.

For the design of the classifiers, the hierarchical classification function is used, in order to improve the prediction of the overlapping class. Fig. 14 represents the descriptive model, which is independent of the algorithm to be used. Taxonomy A groups the healthy and diabetic classes into Diabetes group, considering that the diabetic subject is better identified over the healthy subject in attributes F80, MNF and SampEn in the X axis. In this way, model first seek to recognize the condition of diabetes in the healthy group, to be compared later with

neuropathic instances. On the other hand, in taxonomy B the diabetic class remains in a different node, while the classifier groups the healthy and neuropathic classes in Neuropathy node. This time, it seeks to recognize the pathology in the healthy group, mainly in attributes of the Y axis, which will then be compared with the instances of diabetic subjects. In this case, all attributes are in different quartiles, which improves the classification. Both models are divided by range of age: less or equal to 60 years old and greater than 60 years old. A third taxonomy grouping between diabetics and neuropathic patients does not allow an improvement in the classification, therefore, another model is not considered.

6. Results

6.1. Final dataset validation

A comparison of Path and F80 attributes between real and augmented data is found in Fig. 15 and Fig. 16, in the COP X and COP Y, respectively, where it is visualized that artificial data have more centered distribution with fewer outliers than real values, due to the effect of random selection based on the median. However, on the X axis there is a certain separation between healthy subjects (green) and diabetics (blue), but more prominent with neuropathic subjects (orange). In the case of Y axis, only neuropathic ones (orange) differ from the other classes, because they reach higher ranges of values in the attributes than the other two classes. Fig. 17 shows the pair plot of the final dataset, which includes real and artificial data, with the four attributes of the frequency domain (mean PSD in the LF, MF and HF bands and the F80) and the three attributes based on posture (MeanVel, SampEn and Path). Additionally, two attributes of the DVC dataset are included: sex and age of patients. In addition, axis is added in one column instead of calculating both values for X and Y in separate attributes.

6.2. Classification models

The evaluation metrics in the hierarchical classification are accuracy, error, sensitivity, specificity and Area Under Curve (AUC), considering that each node corresponds to a binary classification. For the validation and testing of the model, precision and recall are used instead of sensitivity and specificity, because the classifier generalizes the taxonomy, turning the problem into a multiclass classification.

Table 6 corresponds to the training of FFN algorithm, which obtained the highest performance of the four used, applied to six cases that differ in the changes of conditions of the model. Each one considers selecting one of the six datasets: taxonomy A (case 1), filtered by age less than or equal to 60 years (case 2), filtered by age greater than 60 years (case 3) and the previous three but considering taxonomy B (case 4), filtered by age less than or equal to 60 years (case 5) and filtered by age greater than 60 years (case 6). A summary of cases is found in Table 5. Training of models is carried out per node, depending on the selected taxonomy: if the taxonomy is A, then first it will classify the healthy — diabetic classes that are in the Diabetes group, at the lowest level, and then go up one step and classify the Diabetes group with separated neuropathic class (Root node). If the taxonomy is B, then the Neuropathy group will be classified with the diabetic class, after classifying between neuropathic — healthy within the Neuropathy group (similar to what occurs in taxonomy A).

Results shown in Table 6 indicate that age-filtered sets perform better than sets that do not divide by age. The model that obtains the best performance is case 3, with taxonomy A and filtered by age over 60 years; with 98% of accuracy in the parent nodes and 100% in the diabetic — healthy classes, followed by case 5, with taxonomy B and filtered by age range less than 60 years. This shows that hierarchical classification improves the identification of overlapping classes, by conditioning the binary comparison between less overlapping classes.

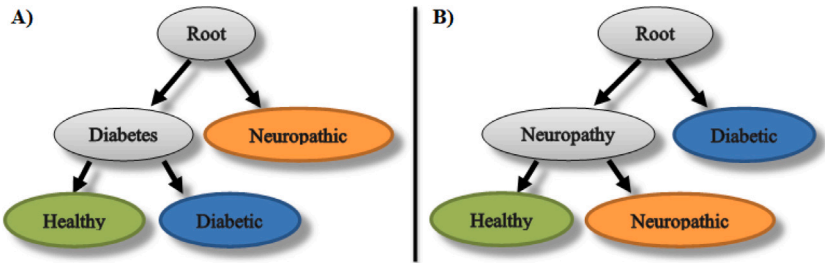


Fig. 14. Descriptive model of taxonomy A and B for hierarchical classifier.

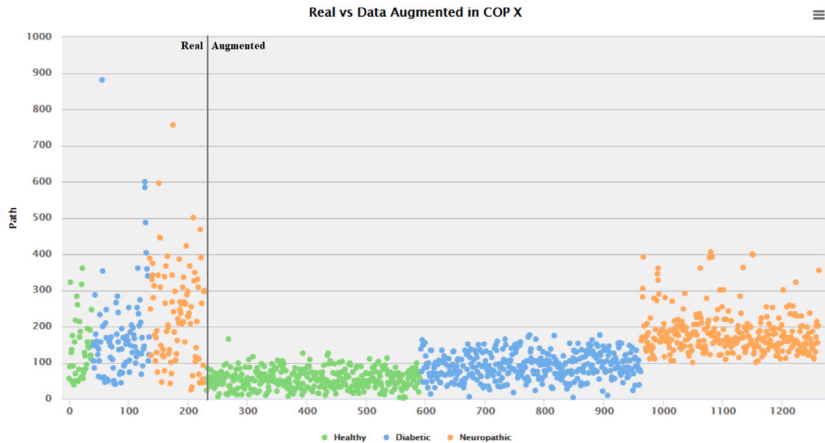


Fig. 15. Real vs augmented data of Path values on COP X in the final dataset.

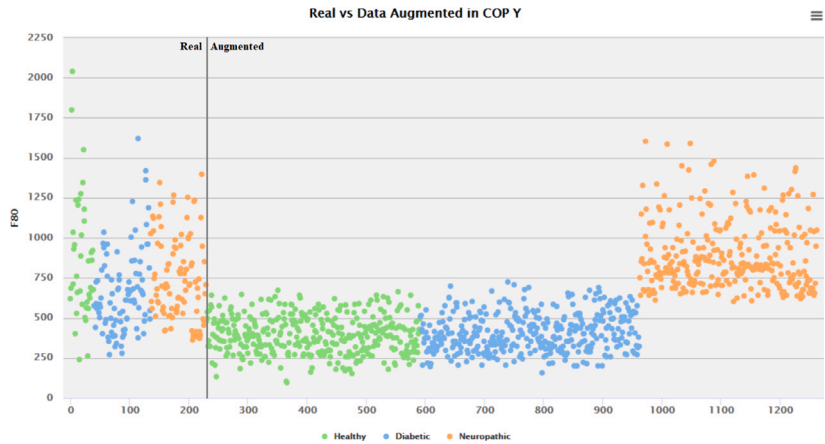


Fig. 16. Real vs data augmented of F80 values on COP Y in the final dataset.

Table 5
Summary of the six cases used for training the classifiers.

Case	Taxonomy	Nodes
1	A	Root: (Diabetes-Neuropathic) Diabetes: (Diabetic-Healthy)
2	A, Age ≤ 60	Root: (Diabetes-Neuropathic) Diabetes: (Diabetic-Healthy)
3	A, Age > 60	Root: (Diabetes-Neuropathic) Diabetes: (Diabetic-Healthy)
4	B	Root: (Neuropathy-Diabetic) Neuropathy: (Neuropathic-Healthy)
5	B, Age ≤ 60	Root: (Neuropathy-Diabetic) Neuropathy: (Neuropathic-Healthy)
6	B, Age > 60	Root: (Neuropathy-Diabetic) Neuropathy: (Neuropathic-Healthy)

Table 6
Results from training hierarchical classification of a FFN algorithm.

Case	Accuracy	Error	Sensitivity	Specificity	AUC
1	93%	7%	90%	94%	0.9
	97%	3%	98%	96%	0.9
2	99.6%	0.4%	99.7%	99.5%	0.9
	97%	3%	97%	97%	0.9
3	98%	2%	97%	98%	0.9
	100%	0%	100%	100%	1
4	93%	7%	88%	95%	0.9
	97%	3%	96%	97%	0.9
5	99.7%	0.3%	99.9%	99.7%	0.9
	97%	3%	97%	97%	0.9
6	98%	2%	97%	98%	0.9
	99.6%	0.4%	99.7%	99%	0.9

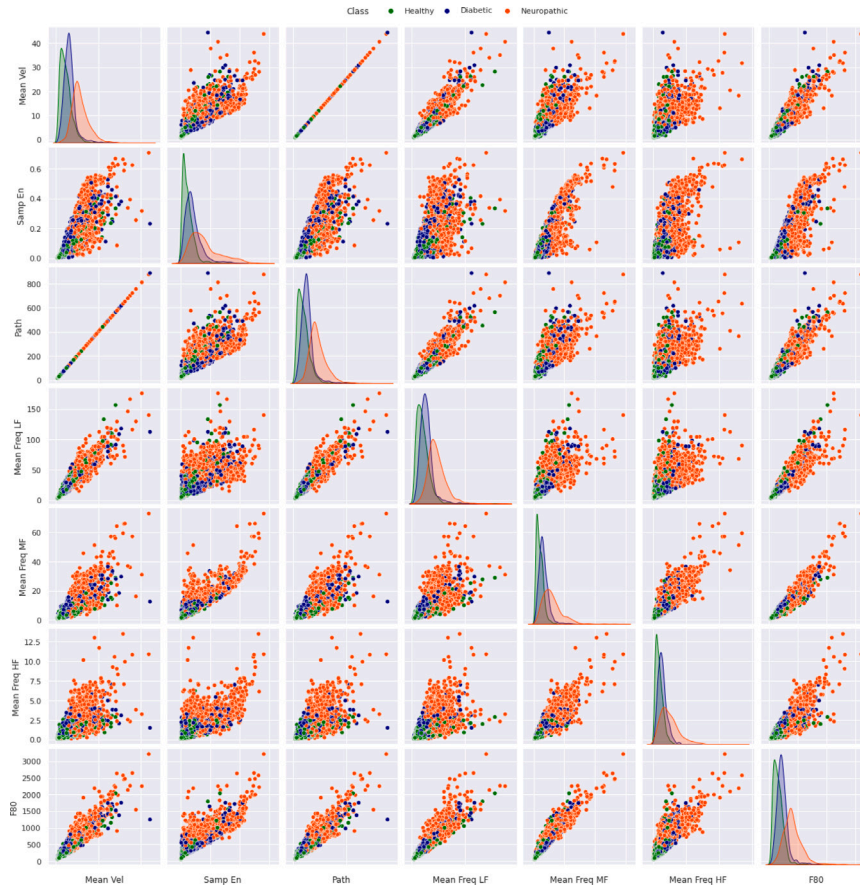


Fig. 17. Pairplot of numeric attributes in the final dataset.

Also, it shows the tendency that features tend to become more complex as the subject turns older.

In the validation and testing set, results correspond to the final classification of new instances that go through the complete taxonomy. The algorithm does not separate the results by levels as it was in the training set. For this reason, new instance will be classified in one of the three corresponding classes, therefore the binary classifier is “converted” to a multiclass classifier. Results show an improvement in the classification of the overlapping classes. The validation set in Table 7 shows the two algorithms that obtained a performance greater than 70% of the F-Score: the FFN and the RF algorithms, the first being the one that achieves a 97% of F-Score, the highest of the six applied cases. The first four results (cases 5, 2, 3 and 6) suggest that selected taxonomy and the separation in age range significantly improve the classification of the subjects. In the case of the RF algorithm, the optimal classification conditions are similar to those of the FFN model, with the lowest performance being those of the dataset that does not consider an age filter. In the case of the K-NN algorithm, it does not exceed 75% of F-Score, followed by the NB algorithm, which does not exceed 65%, therefore, they are not listed in Table 7. In Table 8, results of testing dataset are shown. Again the FFN algorithm achieves a higher performance, surpassing the F-Score of 90% over the RF algorithm, which reaches up to 88% in the best case. In both models, cases 5 (taxonomy B and age less or equal than 60 years old) and 2 (taxonomy A and age less or equal than 60 years old) have the highest performance.

The results of the test set are listed below in Table 9, for the KNN and NB algorithms, which got the lowest F-Score in comparison with FFN network and Random Forest.

Table 7

Validation results of hierarchical classification with FFN and RF algorithms.

Case	Algorithm	Accuracy	Error	Recall	Precision	F-Score
5	FFN	97%	3%	97%	97%	97%
2	FFN	97%	3%	97%	97%	97%
3	FFN	96%	4%	96%	96%	96%
6	FFN	96%	4%	96%	96%	96%
4	FFN	89%	11%	89%	90%	89%
1	FFN	89%	11%	89%	89%	89%
2	RF	87%	13%	87%	88%	87%
5	RF	87%	13%	87%	88%	87%
3	RF	80%	20%	79%	83%	81%
6	RF	80%	20%	79%	83%	81%
4	RF	75%	25%	75%	79%	77%
1	RF	74%	26%	74%	78%	76%

Table 8

Testing results of hierarchical classification with FFN and RF algorithms.

Case	Algorithm	Accuracy	Error	Recall	Precision	F-Score
2	FFN	97%	3%	97%	97%	97%
5	FFN	97%	3%	97%	97%	97%
3	FFN	96%	4%	95%	95%	95%
6	FFN	95%	5%	95%	95%	95%
4	FFN	90%	10%	90%	90%	90%
1	FFN	89%	11%	90%	90%	90%
2	RF	87%	13%	87%	88%	88%
5	RF	87%	13%	87%	88%	88%
3	RF	80%	20%	79%	82%	81%
6	RF	80%	20%	79%	82%	81%
4	RF	71%	29%	71%	74%	73%
1	RF	71%	29%	71%	73%	72%

Table 9
Testing results of hierarchical classification with K-NN and NB algorithms.

Case	Algorithm	Accuracy	Error	Recall	Precision	F-Score
2	K-NN	72%	29%	72%	72%	72%
5	K-NN	72%	29%	72%	72%	72%
1	K-NN	69%	31%	69%	69%	69%
4	K-NN	68%	32%	68%	69%	69%
3	K-NN	66%	34%	66%	67%	66%
6	K-NN	66%	34%	66%	67%	66%
2	NB	63%	37%	63%	67%	65%
5	NB	63%	37%	63%	67%	65%
3	NB	59%	41%	61%	63%	62%
6	NB	59%	41%	61%	63%	62%
4	NB	60%	40%	61%	62%	61%
1	NB	58%	42%	59%	62%	61%

7. Discussion and future work

The characteristics based on PSD show that data have a behavior of increasing tendency, where healthy subjects always remain in low quartiles, diabetics in medium quartiles and neuropathic subjects in medium-high quartiles, as the disease worsens. Frequency domain attributes reach to capture the underlying behaviors, mainly when including distractors during evaluation. The distribution of PSD is associated to which part of the equilibrium system it is intended to alter: by disturbing vision, the PSD shifts from low frequencies to medium frequencies. When including a dual task or unstable surface, PSD shifts from low frequencies towards medium and high frequencies, being more erratic in certain patients. This implies an improvement in the quality of the attributes, considering the limitations of the sampling rate presented by the WBB. On the other hand, the age and the axis are the most important attributes, due to it is verified that both influence those data that are overlapping. For this reason, separating axis allows the differentiation of health condition: COP X between healthy and neuropathic patients, with a certain degree of separation with diabetics; and COP Y between neuropathic and healthy patients. However, the hierarchical classifier optimizer reach to solve this problem, assigning a more structured role when classifying new instances.

Data augmentation makes it possible to compensate for the reduction in instances caused by grouping by the hierarchical classifier. Because no further evidence was found on an ideal number to divide series in similar works, the selection, crossover and mutation parameters were established empirically. The criterion used was that the algorithm generated the smallest number of duplicated time series. In the case of crossover and mutation, the appropriate value to divide the series was 10, but the value to cross was 5 for HG and DG and 4 for DNG. For mutation, 4 was the value that generated the greatest number of unduplicated series. A greater number of divisions only caused the genetic algorithm to take longer to reach 5000 iterations and a high number of duplicates. In the case of using a number less than 10, the execution time of the algorithm was shorter, but the number of duplicates was greater. Nevertheless, given the nature of the genetic algorithm, it allows obtaining new instances that inherit the underlying behaviors of the real data. This is verified in the data exploration, for example, between the Path and F80 median values, where the augmented data ranges in similar quartiles of real data. The difference lies in that real data present more overlapping atypical values, in the three classes. This is presumably due to other characteristics that the patients present (other underlying diseases or health conditions) that are not being captured in the postural evaluation process. Nevertheless, the artificial data maintains the growing trend conditions.

Regarding the results of the machine learning models, only the FFN achieves a 97% of F-Score in the testing set, mainly when dividing by age range. Then the RF model achieves up to 87% F-Score. This implies that neural network model with ReLU activation functions is better than the entropy calculation of decision trees. Regarding the algorithms with lower performance, the K-NN algorithm uses Euclidean

distance to measure the similarity between data instances. However, in cases where there is overlap between classes, this metric may not be discriminative enough to distinguish overlapping classes. As noted by Hastie, Tibshirani and Friedman [58], K-NN can be highly sensitive to the presence of noise and irrelevant features, especially in high-dimensional data sets or when classes are overlapping. On the other hand, NB uses conditional probability to estimate the most probable class given a data instance and assumes conditional independence between features. This simplification can lead to a lack of robustness when there is overlap between class distributions. This shows that, despite the inclusion of frequency domain and posture features, the dispersion between attributes is still complex to adjust. For this reason, K-NN model reach 72% of the F-Score with the dataset divided by age less than or equal 60 years old, in both taxonomies. The lowest performance is for NB, with does not even reach 65% of the F-Score. Therefore, the adjustment of the FFN and RF algorithms must be further studied, with the aim of improving the metrics and discarding the K-NN and NB, due to their limitations with the current data set.

With respect to the six designed use cases, a higher performance in cases 2 (Taxonomy A, Age ≤ 60) and 5 (Taxonomy B, Age ≤ 60) for all four algorithms suggests that, from a certain age range, the distribution of the classes become more complex. By separating adults under 60 years old, the algorithms classify better than when the entire set of participants is included, as occurs with cases 4 (Taxonomy A without separation) and 1 (Taxonomy B without separation). In the work of Remaud, Thuong-Cong and Bilodeau [59] there is evidence that, in seniors over 60 years old, the complexity of dynamic postural control is greater than in young adults, because seniors adopt a mode of reactive control to postural perturbations. Other antecedents in the review by Osoba, Rao and others [25] suggest that aging impairs sensory systems, reducing the ability to balance, which results in greater overlap of data. Therefore, an alternative would be to look for other distractors that allow increasing the differentiating features of each individual, mainly for adults over 60 years of age, where the worsening of the disease is reflected in the superposition of the data of the diabetic and the neuropathic.

In [60], convolutional neural networks (CNN) and long-term short memory (LTSM) models were used for feature extraction and prediction of COVID-19 hot-spots from time-series data. In this work, deep learning models (e.g. LSTM, CNN) were not used mainly because the number of data available, even after data augmentation, and the huge number of parameters required for training this kind of models. As a future work, deep neural network models based on transfer learning would be explored and compared with the models used in this work. In addition, hierarchical classification models provide graphical and understandable models in comparison with Deep learning models, which are essentially black boxes from the point of view of a domain expert.

As future work, the alternative of including other evaluation techniques may be considered, in order to have more information on the variables that aggravate diabetes and neuropathy.

A repository on GitHub is available to verify data at: <https://github.com/vidcoo/GenAlgorithm>.

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CRediT authorship contribution statement

Claudio Meneses Villegas: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Jorge Littin Curinao:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **David Coo Aqueveque:** Writing – original draft, Visualization, Validation, Investigation, Formal analysis. **Juan Guerrero-Henríquez:** Writing – review & editing, Validation, Methodology, Formal analysis. **Martín Vargas Matamala:** Writing – review & editing, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared a link to the code/data in the manuscript.

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